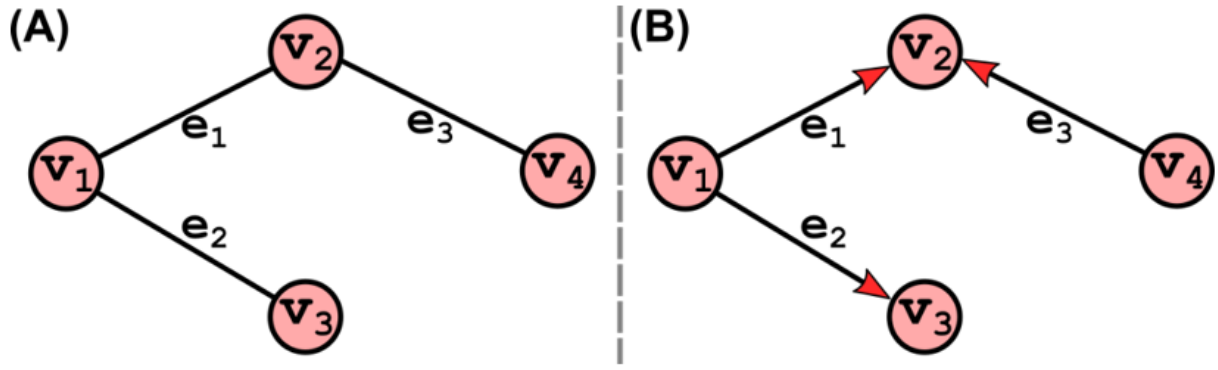


Figure 14.4 An undirected graph (A) versus a directed graph with the same vertices (B). The two graphs are not equivalent, in particular the latter can't be transformed into an equivalent undirected graph.

The vice versa is usually not true, and many directed graphs can't be transformed into their undirected isomorphic counterpart[3].

So, unless the application context suggests otherwise, representing directed graphs is the least restricting choice.

It's also worth noting that, if the adjacency matrix representation is used, directed graphs can be represented using only half the matrix, because they always have a symmetrical adjacency matrix: $A[u,v] = A[v,u]$ for every pair of vertices u, v.

14.2.2 Connected

A graph is *connected* if, given any pair of its vertices (u, v), there is a sequence of vertices $u, (w_1, \ldots, w_K,), v$, with $k \geq 0$, such that there is an edge between any two adjacent vertices in the sequence.